<u>Mark scheme and solutions</u>

Paper 1: Pure maths

1a. Square numbers are (0), 1, 4, 9, 16, 25, 36, 49, 64, 81, (100) [1 mark]
None of these end in 3, and adding multiples of 10 to a number does not affect its units digit [1 mark]
So no square ends in 3 [1 mark]

1b. When n = 4, u_n = 1 + 2 + 6 + 24 = 33 [1 mark]
For n > 4, all numbers being added are multiples of 10 [1 mark]
since 5! contains 5 and 2 as factors [1 mark]
So by the result in part a) there are no square numbers when n ≥ 4 [1 mark]

2a. Notice that $t^4 + 1$ is 2 more than $t^4 - 1$, which divides $t^2 - 1$ (difference of two squares) so

$$\frac{t^4+1}{t^2-1} = \frac{t^4-1+2}{t^2-1} = \frac{(t^2-1)(t^2+1)+2}{t^2-1} = t^2+1+\frac{2}{t^2-1}$$

[2 marks] *(could also have used polynomial long division to get this step)*

Then using partial fractions (by inspection)

$$\frac{2}{t^2-1} = \frac{1}{t-1} - \frac{1}{t+1}$$ [1 mark]

So the final answer is

$$t^2+1+\frac{1}{t-1}-\frac{1}{t+1}$$ [1 mark]

2b. Start by expanding the LHS then using various trig identities:

$= \cos^2 x + \cos^2 y + 2 \cos x \cos y + \sin^2 x + \sin^2 y - 2 \sin x \cos y$ [1 mark]

$= 2 + 2 \cos x \cos y - 2 \sin x \sin y$ [1 mark]

$= 2 + 2 \cos(x + y)$ [1 mark]

Let $\theta = (x + y)/2 \rightarrow x + y = 2\theta$ and use the double angle formula:

$= 2 + 2 \cos 2\theta$

$= 2 + 4 \cos^2 \theta - 2$ [1 mark, or without substitution]

Unsubstitute θ:

$= 4 \cos^2 (x + y)/2$ (= RHS). [1 mark]

3a. $y = \sin^{-1} x \rightarrow x = \sin y$ [1 mark]

Differentiate both sides with respect to x: $1 = \cos y * dy/dx$ [1 mark]

Rearrange: $dy/dx = 1 / \cos y$ [1 mark]

Put y back in: $dy/dx = 1 / \cos \sin^{-1} x$

Use identity $\cos x = \pm \sqrt{(1 - \sin^2 x)}$

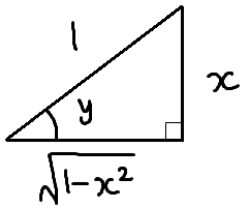
$\rightarrow dy/dx = 1 / \pm \sqrt{(1 - \sin^2 \sin^{-1} x)}$

$\rightarrow dy/dx = 1 / \pm \sqrt{(1 - x^2)}$ [1 mark]

$y = \sin^{-1} x$ is an increasing function so gradient is always positive $\rightarrow$ reject -ve

$\rightarrow dy/dx = 1 / \sqrt{(1 - x^2)}$ [1 mark]

(Could have also drawn a simple right-triangle to avoid the identity:



so $\cos y = 1 / \sqrt{(1 - x^2)}$ directly. Still must justify rejecting -ve solution.)

3b. i) Starting with

$$f(x) = x\sqrt{a^2 - x^2} + a^2 \sin^{-1} \frac{x}{a}$$

The first term uses the product rule. To differentiate $\sqrt{(a^2 - x^2)}$, write as $(a^2 - x^2)^{1/2}$ and use chain rule, which gives $\frac{1}{2}(a^2 - x^2)^{-1/2}(-2x)$.

The second term uses the previous result, with a factor of 1/a required from the chain rule. This produces

$$f'(x) = \sqrt{a^2 - x^2} - \frac{x^2}{\sqrt{a^2 - x^2}} + \frac{a}{\sqrt{1 - \left(\frac{x}{a}\right)^2}}$$

[4 marks; 2 for first two terms, 2 for last term]

Multiply the top and bottom of the third term by $a = \sqrt{a^2}$:

$$f'(x) = \sqrt{a^2 - x^2} - \frac{x^2}{\sqrt{a^2 - x^2}} + \frac{a^2}{\sqrt{a^2 - x^2}}$$ [1 mark]

Factor out $1/\sqrt{(a^2 - x^2)}$ from the second and third terms:

$$f'(x) = \sqrt{a^2 - x^2} + \frac{a^2 - x^2}{\sqrt{a^2 - x^2}} = \sqrt{a^2 - x^2} + \sqrt{a^2 - x^2} = 2\sqrt{a^2 - x^2}.$$

[1 mark]

3b. ii) Equation of circle: $x^2 + y^2 = a^2 \rightarrow y = \sqrt{(a^2 - x^2)}$ [1 mark]
(positive solution only since semi-circle is above x-axis)
Region is symmetric about y-axis so the area can be written as

$$\text{area} = 2\int_0^{a/2} \sqrt{a^2 - x^2}\ \mathrm{d}x = \int_0^{a/2} 2\sqrt{a^2 - x^2}\ \mathrm{d}x$$ [1 mark]

This is the same as the result in 3b.i), so

$$\text{area} = \int_0^{a/2} f'(x)\ \mathrm{d}x = f\left(\frac{a}{2}\right) - f(0)$$ [1 mark]

Putting in x = 0 into f(x), we find f(0) = 0, so the area is f(a/2).
[1 mark]

3b. iii) Area of semicircle = $\frac{1}{2}\pi a^2$ [1 mark]

Area of shaded region = f(a/2) = (a/2) * $\sqrt{(3a^2/4)} + a^2 \sin^{-1} \frac{1}{2}$

= $\sqrt{3}/4\ a^2 + 1/6\ \pi a^2$ [1 mark]

Ratio = $(\sqrt{3}/4 + 1/6\ \pi) / (1/2\ \pi)$

$= \frac{\sqrt{3}}{2\pi} + \frac{1}{3}.$ [1 mark]

4a. f(x) is undefined when either the input to the log is zero, or there is a division by zero in the fraction (since log(infinity) diverges).

ln(0) occurs when: 1 + sin x = 0 → sin x = -1 → x = 3π/2, -π/2 [1 mark]
Division by zero occurs when: 1 - sin x = 0 → sin x = 1 → x = π/2, -3π/2 [1 mark]

The set is { -3π/2, -π/2, π/2, 3π/2 } = { ±π/2, ±3π/2 } [1 mark].

4b. We need to turn 1 and sin x into sec x and tan x. The easiest way to do this is to divide top and bottom by cos x:

$$f(x) = \frac{1}{2}\ln\frac{(1+\sin x)\div\cos x}{(1-\sin x)\div\cos x} = \frac{1}{2}\ln\frac{\sec x+\tan x}{\sec x-\tan x}$$

[2 marks; 1 for choosing to divide by cos x, 1 for getting the correct form]
Multiply top and bottom by the conjugate to create difference of squares:

$$= \frac{1}{2}\ln\frac{(\sec x+\tan x)\cdot(\sec x+\tan x)}{(\sec x-\tan x)\cdot(\sec x+\tan x)} = \frac{1}{2}\ln\frac{(\sec x+\tan x)^2}{\sec^2 x-\tan^2 x}$$ [2 marks]

Use the identity $\sec^2 x = 1 + \tan^2 x \rightarrow \sec^2 x - \tan^2 x = 1$:

$$= \frac{1}{2}\ln(\sec x+\tan x)^2$$ [1 mark]

Use the log identity, $\log x^a = a \log x$. The 1/2 cancels with the square, take the modulus to preserve the domain:

$= \ln|\sec x+\tan x|.$ [1 mark]

4c. i) Starting with

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln|\sec x+\tan x|,$$

use chain rule: d/dx ln|x| = 1/x (no modulus), d/dx tan x = $\sec^2 x$

[1 mark for recalling and applying both these] and

d/dx sec x = d/dx 1 / cos x = sin x / $\cos^2 x$ = sec x tan x [1 mark], so

$$=\frac{\sec x\tan x+\sec^2 x}{\sec x+\tan x}=\frac{\sec x(\sec x+\tan x)}{\sec x+\tan x}=\sec x.$$ [1 mark]

4c. ii) Differentiate g(x) using the product rule:

g'(x) = -cos x * f(x) + (1 - sin x) * f'(x)

= -cos x ln|sec x + tan x| + (1 - sin x) sec x [1 mark]

Transform x to x + π/2. This represents a shift of π/2 to the left. We can visualise / quickly sketch the graphs to see that cos(x + π/2) = -sin x, sec(π + π/2) = -csc x, tan(x + π/2) = -cot x and sin(x + π/2) = cos x [4 marks]. Putting these in,

g'(x + π/2) = sin x ln|-csc x - cot x| - (1 - cos x) csc x

= (cos x - 1) csc x + sin x ln|csc x + cot x| [1 mark]

Now applying small-angle approximations,

sin x ≈ x, cos x ≈ 1 - $\frac{1}{2}x^2$, csc x = 1 / sin x ≈ cot x = 1 / tan x ≈ 1 / x,

g'(x + π/2) ≈ (1 - $\frac{1}{2}x^2$ - 1) (1 / x) + x ln|1 / x + 1 / x| [2 marks]

≈ -$\frac{1}{2}$ x + x ln|2 / x|. [1 mark]

4c. iii) g(x) is undefined at x = π/2 (since from part a, f(x) was also undefined there). [1 mark]

5a. Multiplying through by dθ/dt:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t}\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{L}\frac{\mathrm{d}\theta}{\mathrm{d}t}\sin\theta = 0$$ [1 mark]

In the first term, we see a function (dθ/dt) times its derivative ($d^2\theta/dt^2$). This seems similar to the chain rule. With a bit of thought we can see that if we differentiate $(d\theta/dt)^2$ we get 2 (dθ/dt) ($d^2\theta/dt^2$).

Following a similar process for the second term we can deduce that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2\right) + \frac{g}{L}\frac{\mathrm{d}}{\mathrm{d}t}\left(-\cos\theta\right) = 0$$ [2 marks]

Now integrate both sides with respect to *t*, we eliminate the d/dt and get a constant on the RHS (integral of 0 is 0t + *C*)

$$\frac{1}{2}\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 - \frac{g}{L}\cos\theta = C$$ [1 mark]

5b. Initial conditions: $\theta = \theta_0$ and dθ/dt = 0 (since it was released from rest i.e. speed = angular speed = 0)

Putting this in to the LHS:
$\frac{1}{2} * 0^2 - g/L * \cos\theta_0 = C \rightarrow C = -g/L \cos\theta_0$ [1 mark]

Put this back in, rearrange and factorise:

$$\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = \frac{2g}{L}\left(\cos\theta - \cos\theta_0\right)$$ [1 mark]

5c. Applying separation of variables:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \sqrt{\frac{2g}{L}(\cos\theta - \cos\theta_0)}$$ [1 mark, no need to justify removing - sign]

$$\sqrt{\frac{L}{2g}}\frac{1}{\sqrt{\cos\theta - \cos\theta_0}}\,\mathrm{d}\theta = \mathrm{d}t$$ [1 mark, allow constant factor still inside]

Thinking about the motion of the pendulum, the time period will be the time taken to move through a complete oscillation: θ goes from θ_0, down through 0, up to $-\theta_0$, back down through 0 and up to θ_0 again. Since the times taken to do each of these is equal, we can say that $\frac{1}{4}$ of the time period is when the pendulum returns from 0 to θ_0 (to make the signs easiest).

Applying the integration, therefore

$$\sqrt{\frac{L}{2g}}\int_0^{\theta_0}\frac{1}{\sqrt{\cos\theta - \cos\theta_0}}\,\mathrm{d}\theta = \int_0^{T/4}\mathrm{d}t = \frac{T}{4}$$ [1 mark]

To use the substitution given in the question, we now need to convert cosines to half-angle sines. This can be done using the double angle formula:
$\cos 2x = 1 - 2\sin^2 x$: let $x = \frac{1}{2}\theta$ and $\frac{1}{2}\theta_0$ (separately):
$\rightarrow \cos\theta = 1 - 2\sin^2\frac{1}{2}\theta$ and $\cos\theta_0 = 1 - 2\sin^2\frac{1}{2}\theta_0$:

$$\frac{T}{4} = \sqrt{\frac{L}{2g}}\int_0^{\theta_0}\frac{1}{\sqrt{1 - 2\sin^2\frac{1}{2}\theta - 1 + 2\sin^2\frac{1}{2}\theta_0}}\,\mathrm{d}\theta$$

$$T = 4\sqrt{\frac{L}{2g}}\int_0^{\theta_0}\frac{1}{\sqrt{2\sin^2\frac{1}{2}\theta_0 - 2\sin^2\frac{1}{2}\theta}}\,\mathrm{d}\theta$$ [2 marks]

Now consider the given substitution, $\sin \frac{1}{2} \theta = \sin \frac{1}{2} \theta_0 \sin x$. Differentiating implicitly with respect to x:

$\rightarrow \frac{1}{2} \cos \frac{1}{2} \theta * d\theta/dx = \sin \frac{1}{2} \theta_0 \cos x$

$\rightarrow d\theta = (2 \sin \frac{1}{2} \theta_0 \cos x) / (\cos \frac{1}{2} \theta)\, dx$ [1 mark]

The new limits of the integral are:

$0 = \sin x \rightarrow x = 0$ and $\sin \frac{1}{2} \theta_0 = \sin \frac{1}{2} \theta_0 \sin x \rightarrow \sin x = 1 \rightarrow x = \pi/2$.

Applying these substitutions, we get

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2 \sin \frac{1}{2}\theta_0 \cos x}{\cos \frac{1}{2}\theta \sqrt{2 \sin^2 \frac{1}{2}\theta_0 - 2 \sin^2 \frac{1}{2}\theta_0 \sin^2 x}}\, \mathrm{d}x$$

[1 mark]

Factor out $\sin \frac{1}{2} \theta_0$ from inside the square root, it cancels with the top:

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2 \cos x}{\cos \frac{1}{2}\theta \sqrt{2 - 2 \sin^2 x}}\, \mathrm{d}x$$

[1 mark]

We now need to get rid of the θ term. Using the Pythagorean trig identity,

$\cos^2 \frac{1}{2} \theta = 1 - \sin^2 \frac{1}{2} \theta$

$= 1 - \sin^2 \frac{1}{2} \theta_0 \sin^2 x$ (using the substitution again)

Putting this under the square root,

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2 \cos x}{\sqrt{(1 - \sin^2 \frac{1}{2}\theta_0 \sin^2 x)(2 - 2 \sin^2 x)}}\, \mathrm{d}x$$

[1 mark]

Expand the brackets:

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2 \cos x}{\sqrt{2 - 2 \sin^2 x - 2 \sin^2 \frac{1}{2}\theta_0 \sin^2 x + 2 \sin^4 x \sin^2 \frac{1}{2}\theta_0}}\, \mathrm{d}x$$

Replace $2 - 2\sin^2 x = 2\cos^2 x$ and $2\sin^4 x = 2\sin^2 x\,(1 - \cos^2 x)$:

$$T = 4\sqrt{\frac{L}{2g}}\int_0^{\frac{1}{2}\pi} \frac{2\cos x}{\sqrt{2\cos^2 x - 2\sin^2 x\cos^2 x\sin^2\frac{1}{2}\theta_0}}\,\mathrm{d}x$$

[2 marks]

Finally, factor out $\sqrt{2}\cos x$ from under the square root, it cancels with the top and the $\sqrt{2}$ cancels with the factor at the front - we get

$$T = 4\sqrt{\frac{L}{g}}\int_0^{\frac{1}{2}\pi} \frac{1}{\sqrt{1 - \sin^2 x\sin^2\frac{1}{2}\theta_0}}\,\mathrm{d}x$$

[1 mark]

5d. i) Step size = 90/4 = 22.5° (π/8 radians).

The constant factor is $\sin^2 \frac{1}{2}\theta_0 = \sin^2 30 = \frac{1}{4}$ (to save time, use this)

Using trapezium rule on the integral part:

$I = \pi/16 * (1 + 2 * (\frac{1}{\sqrt{1-\frac{1}{4}sin^2\frac{\pi}{8}}} + \frac{1}{\sqrt{1-\frac{1}{4}sin^2\frac{\pi}{4}}} + \frac{1}{\sqrt{1-\frac{1}{4}sin^2\frac{3\pi}{8}}}) + \sqrt{\frac{4}{3}})$.

I = 1.685750355 [3 marks]

→ T = 4 * √(0.5 / 9.81) * 1.685750355 = 1.522312276 [1 mark]

5d. ii) Exact value = 4 * √(0.5 / 9.81) * 1.685750355 = 1.522312276 [1 mark]

(appears to be) exactly correct [1 mark]

Accuracy is extremely high / exact because the function is (almost) equally concave up as concave down on the interval so the error terms (almost) cancel out [1 mark]

Not the function is a straight line

(In fact the answer is not quite exact, the error is about 3×10^{-10}. However, since the calculator only displays 10 significant figures, the answer appears to match exactly.)

Mechanics

6a. Let mass be m, the weight acts at the midpoint (d metres from P).
Taking moments about P, with clockwise as positive:

$10 * 2000g = 10 * 400g$ + moment due to crane's weight [1 mark]
moment due to crane's weight = $16000g$ [1 mark]
$16000g = m * g * (40/2) \rightarrow m = 800$ kg [1 mark]

b. Additional moment due to moving load = $x * 400g$ [1 mark]
Additional moment due to moving counterweight = $y * 2000g$ [1 mark]
Moments must balance: $400gx = 2000gy \rightarrow x = 5y \rightarrow x : y = 5 : 1$ [1 mark]

(Alternative method: take moments again in the new positions:
$(10 + y) * 2000g = (10 + x) * 400g + 16000g$ [1 mark]
$20000g + 200gy = 4000g + 400gx + 16000g$ [1 mark]
$400gx = 2000gy \rightarrow x = 5y \rightarrow x : y = 5 : 1$ [1 mark]*)*

7a. D = distance = area under speed-time graph.
Shape is a trapezium, so use $A = \frac{1}{2} h(a + b)$:
$\rightarrow D = \frac{1}{2} V(T + (T - 2T/6))$ [1 mark]
$\rightarrow D = (5/6) VT$ [1 mark]
Next, eliminate V. Since acceleration = change in speed / time,
$\rightarrow m = V/(t_1) = V/(T/6) = 6 * V/T$ [1 mark]
$\rightarrow V = (1/6) mT$ [1 mark]
Substituting this back in,
$\rightarrow D = (5/36) mT^2$ [1 mark]

7b. i) The areas under both velocity-time graphs are equal since D is constant. Let U be the new top speed. Let kT be the time spent accelerating (i.e. new t_1 is kT).

Time spent at speed U = (7/9) T - 2kT. [1 mark]
Area under new graph: D = 2 * (1/2 * kT * U) + [(7/9) T - 2kT] * U [1 mark]
= kTU + (7/9 - 2k) TU = (7/9 - k) TU [1 mark]

Equate area with previous result: D = (5/36) mT^2
→ (5/36) mT^2 = (7/9 - k) TU [1 mark]

Eliminating U, use m = U / (kT) → U = mkT [1 mark]
→ (5/36) mT^2 = (7/9 - k) mkT^2 [1 mark]
→ 5/36 = (7/9 - k) k
→ k^2 - (7/9) k + (5/36) = 0 [1 mark]
→ k = 1/2, k = 5/18 [1 mark]

Since k is the fraction of time, we must have 2k ≤ 7/9 since otherwise the total time would not be within (7/9)T. So reject k = $\frac{1}{2}$.
→ time spent accelerating = (5/18) T. [1 mark]

7b. ii) Old top speed = V = (1/6) mT [1 mark]
New top speed = U = mkT = (5/18) mT [1 mark]
Fractional change = (5/18) / (1/6) = 5/3
→ Percentage increase = 100% * (5/3 - 1) = 66.7 % [1 mark]

8a. Using SUVAT in the direction perpendicular to the slope:

$s = 0$ (starts and ends on the slope)

$u = V_n = V \sin\theta$ (resolve velocity vector) [1 mark]

$v = ?$

$a = -g \cos\theta$ (resolve gravitational acceleration vector) [1 mark]

$t = ?$

(usually we have $v = -u$, and in this case it is, but in general when the start and end heights are different, this is not true!)

Using the equation $s = ut + \frac{1}{2} at^2$:

$0 = Vt \sin\theta - \frac{1}{2} gt^2 \cos\theta$ [1 mark]

$0 = t(V \sin\theta - \frac{1}{2} gt \cos\theta)$

Reject $t = 0 \rightarrow V \sin\theta = \frac{1}{2} gt \cos\theta$

$\rightarrow t = 2V \sin\theta / (g \cos\theta) = 2V \tan\theta / g$ [1 mark]

8b. Using SUVAT in the direction parallel to the slope:

$s = L = ?$

$u = V_a = V \cos\theta$ (resolve velocity vector)

$v = ?$ *(again, we cannot say it is equal to u since heights are different)*

$a = g \sin\theta$ (resolve gravitational acceleration vector)

$t = 2V \tan\theta / g$ (from part a) [1 mark for all]

Using the equation $s = ut + \frac{1}{2} at^2$,

$L = (V \cos\theta)(2V \tan\theta / g) + \frac{1}{2} (g \sin\theta)(4V^2 \tan^2\theta / g^2)$ [2 marks]

$L = 2V^2 \sin\theta / g + 2V^2 \sin\theta \tan^2\theta / g$ [1 mark]

$L = 2V^2 \sin\theta / g * (1 + \tan^2\theta)$

$L = 2V^2 \sin\theta / g * \sec^2\theta$ [1 mark]

$L = 2V^2 \sin\theta / (g \cos^2\theta)$ *or* $L = 2V^2 \sec\theta \tan\theta / g$ [1 mark]

8c. Using SUVAT in the direction parallel to the slope:

$s = ?$, $u = V \cos \theta$, $v = ?$, $a = g \sin \theta$ [1 mark]

The place where the distance from the slope is maximum occurs at half the time taken to fall back down → $t = (2V \tan \theta / g) / 2 = V \tan \theta / g$. [1 mark]

Using the equation $s = ut + \frac{1}{2} at^2$,

$s = (V \cos \theta)(V \tan \theta / g) + \frac{1}{2} (g \sin \theta)(V^2 \tan^2 \theta / g^2)$ [2 marks]

$s = V^2 \sin \theta / g + \frac{1}{2} V^2 \sin \theta \tan^2 \theta / g$ [1 mark]

$s = V^2 \sin \theta / g * (1 + \frac{1}{2} \tan^2 \theta)$. [1 mark]

(Alternative methods: find s(t) perpendicular to the slope and differentiate to find the maximum, or find s(t) in the vertical direction and differentiate and set the gradient equal to tan θ)

Statistics

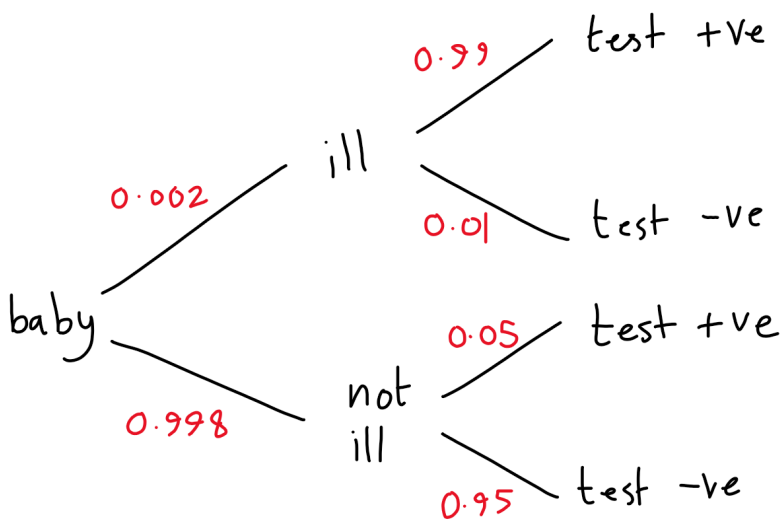
9a. H_0: the true mean of X has not increased / $\mu = 252$ ml
H_1: the true mean of X has increased / $\mu > 252$ ml [1 mark]
Under H_0, $X \sim N(252, 2.5^2) = N(252, 6.25)$
Sample variance = 6.25 / 5 = 1.25 (or $\bar{X} \sim N(252, 1.25)$) [1 mark]
$P(\bar{X} > 255) = 0.0036$ [1 mark]
Since $0.0036 < 0.01$, we reject H_0. [1 mark]. There is sufficient evidence to suggest the mean coffee dispensed is more than 252 ml [1 mark].

(Alternative method: use inverse normal distribution and z-score.)

9b. Use inverse normal distribution with Area = 0.99 (in left-tail) or Area = 0.01 (in right-tail), $\sigma = \sqrt{1.25} = 1.118$ and $\mu = 252$:
→ Critical value: $x = 254.6$.
→ Critical region: $\bar{X} > 254.6$ [1 mark]

(Note: in the non-graphical calculators, left-tail is always assumed and cannot be changed.)

10a. [1 mark for 0.002 and 0.998; 1 mark for the other correct values]



10b. i) (0.002 * 0.99) + (0.998 * 0.05) = 0.0519 [2 marks]

ii) (0.002 * 0.99) + (0.998 * 0.95) = 0.9501 [2 marks]

10c. i) P(ill | test +ve) = P(ill *and* test +ve) / P(test +ve)
= (0.002 * 0.99) / 0.0519 [2 marks]
= 0.0382 (less than 0.05) [1 mark]

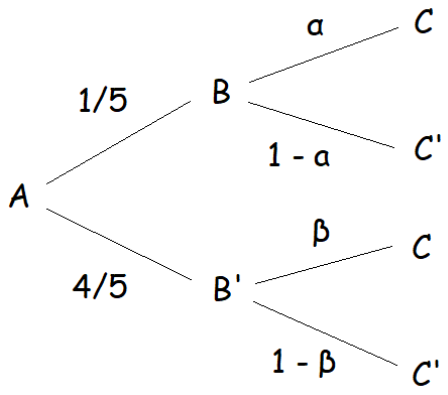
ii) Since the illness is so rare, there are sufficiently many healthy babies that although the false positive rate is low (0.05), this amount still massively outnumbers those who do have the illness and test positive.

[1 mark; anything that says it is due to the massive difference in numbers of the two groups / the fact that the illness is rare.]

10d. i) New P(test +ve) = 1 - new P(test -ve)
= 1 - ((0.998 * p) + (0.002 * 0.01)) [1 mark]
Using the conditional probability formula as in part c.i),
0.50 = (0.002 * 0.99) / (1 - ((0.998 * p) + (0.002 * 0.01))) [1 mark]
Solve for *p*:
p = 0.9980 (99.8%) [1 mark].

ii) Each baby that tests positive could be screened again / multiple times. [1 mark]

11a. i) The tree diagram is:

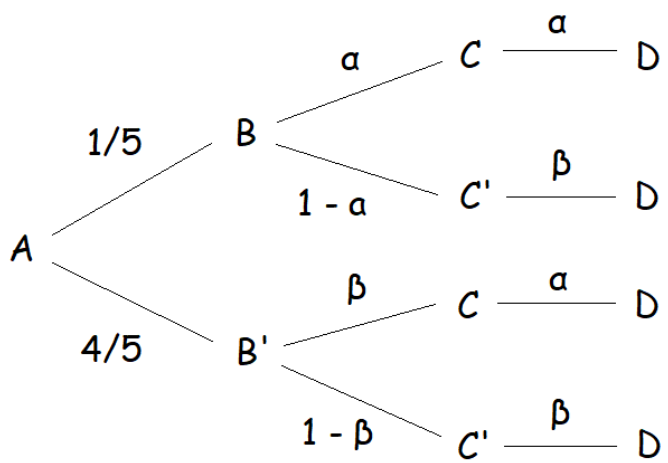


[2 marks]

$P(C) = \alpha/5 + 4\beta/5$

$P(C) = 1/5\ (\alpha + 4\beta)$ [1 mark]

11a. ii) Extending the diagram (branches leading to D' are left out)



[2 marks]

$P(D) = 1/5\ \alpha^2 + 1/5\ (1 - \alpha)\beta + 4/5\ \alpha\beta + 4/5\ (1 - \beta)\beta$

Putting in the given values of α and β,

$P(D) = 0.3295$ [1 mark]

11b. i) Using the tree diagram in part 11a.ii),

$P(X \mid Y) = P(D \mid B) = \alpha^2 + (1 - \alpha)\beta$

$P(X \mid Y') = P(D \mid B') = \alpha\beta + (1 - \beta)\beta$ [3 marks for these]

Subtracting,

$P(X \mid Y) - P(X \mid Y') = \alpha^2 + (1 - \alpha)\beta - \alpha\beta - (1 - \beta)\beta$ [1 mark]

$= \alpha^2 + \beta - \alpha\beta - \alpha\beta - \beta + \beta^2$

$= \alpha^2 - 2\alpha\beta + \beta^2$ [1 mark]

$= (\alpha - \beta)^2$ [1 mark]

11b. ii) If $\alpha = \beta$ then $P(X \mid Y) = P(X \mid Y')$ [1 mark]

Therefore X is independent of Y;

The probability of a neuron in layer D activating is independent of whether or not a neuron in layer B activated. [1 mark]

11c. i) The probability is fixed at 1/5:

$N_B \sim B(1000, 0.2)$ [1 mark]

However, since the number of samples is very large (1000 >> 30), the normal approximation is valid, so:

variance = $\sigma^2 = npq = 1000 * 0.2 * 0.8 = 160$ [1 mark]

$N_B \sim N(200, 160)$ [1 mark]

11c. ii) $\overline{N_B}$ can be modelled with a normal distribution because N_B also had a normal distribution. [1 mark]

$\overline{N_D}$ cannot be modelled with a normal distribution because the distribution of N_D is unknown [1 mark], and the number of trials is not large enough for the central limit theorem to be valid. [1 mark]